

SEBASTIAN MONZÓN

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EDUCATION

05/2023

B.S. Computer Science, Minor in Business

University of Massachusetts Amherst (Honors College) — Amherst, MA

- Thesis Title: *Towards Generative Adversarial Audio Synthesis via Latent Interpolation Between Desired Timbral Qualities*
- GPA: 3.68/4.0

AWARDS

NSF Fellowship (CSGrad4US)

- **\$159,000** over 3 years for a 5-year PhD fellowship period (Starting Fall 2026).

Scholarships

- Melvin Howard Scholarship, Ruth M. White Scholarship, UMass Amherst Dean's Award

HackUMass IX Hackathon Awards

- Won a finalist award and 'Best use of Google Cloud' for a Spotify song queue voting app.

CONFERENCES & PRESENTATIONS

Atmospheric Water Extraction with AI-optimized Music as the Energy-Efficient Alternative to Heat
2025 Material Research Society Fall Meeting, Boston (12/2025)

- Presented a generative AI framework for improving hydrogel water extraction with music.

Music Generation with Stable Diffusion Inpainting

Dept. of Multifunctional Metamaterials (META) at MIT (04/2025)

- Presented a Mel-spectrogram inpainting framework for music generation using Stable Diffusion to Dr. Svetlana Boriskina's cohort.

PUBLICATIONS IN PREPARATION

An AI-Driven Acoustic Symphony: Optimized Hydrogel Dewatering Leveraging a Diffusion-model for Music Spectrogram Inpainting

Shuvo, I., Monzón, S., Smith, G., Flores, R., Diaz-Marin, C., Gu, R., Nagle, S., Boriskina, V., Boriskina, S.

- Designed and tested an AI framework for improved hydrogel air-water extraction using music generated via Mel-spectrogram Stable Diffusion (Riffusion).
- Analyzed hydrogel response to the GTZAN music dataset to inversely design optimal waveforms with optimal frequency ranges.
- Experimented with soft prompting, kernel ridge, SHAP, and other ML methods for inverse design.

A Wearable Waterproof System for Long-Range Health Monitoring and Intelligent Distress Signaling

Shuvo, I., Monzón, S., Boriskina, S.

- Designed, 3D printed, and coded an embedded swim-cap device with long-range radio (ESP32 w/ LoRa), IMU, and pulse sensor for real-time machine learning interpretation of swimmer movement and biomedical data.
- Programmed to transmit, display, and record movement, heart rate, and heart rate variability (HRV) data in real-time with a LoRa receiver.
- Collected live lap-swim data from a swimmer for machine learning analysis.

Musically Interpreting a Simple Generative Transformer

Cosme-Clifford, N., Symons, J., Kapoor, K., Monzón, S., Wm. White, C.

- Explored the aptitudes and failures of a GPT-style model with self-attention, trained on choral-style hymn MIDI files.
- Analyzed the learned latent representations for musical recurrence and chord probabilities.
- Investigated existing self-attention methods for music transformers.

ADDITIONAL RESEARCH EXPERIENCE

MAGE: Motion-to-Audio Generative autoEncoder (06/2025 - Present)

A multi-modal variational auto-encoder (MMVAE) for hand-gesture to audio synthesis

- Incorporated LSTM (Long-Short Term Memory) and 1D Causal TCN (Temporal Convolutional Network) blocks for deep learning of sequential hand position data to BRAVE (Bravely Realtime Audio Variational autoEncoder) latent space mapping.
- Captured high FPS video and audio data to encode hand gestures for accurate positioning and decode to corresponding professional conga drum sounds.
- Applied resampling and peak detection for dataset temporal alignment.
- Used libraries including OpenCV, MediaPipe, and PyTorch.

Music-Emotion Response Study (08/2025 - Present)

Exploring human perceptual responses to music features in collaboration with my undergraduate honors thesis advisor.

- Composed two pop music pieces with fixed chord progressions across and variation across pitch, mode, and rhythmic speed.
- Collaborated with musicologists and data scientists on defining experimental test setup and survey metrics to minimize confounding variables.
- Plan to use EKG, EEG, and GSR to measure the physiological responses of study participants.

Towards Generative Adversarial Audio Synthesis via Latent Interpolation Between Desired Timbral Qualities (05/2023)

My senior year honors thesis exploring Generative Adversarial Networks (GANs) and multi-class classification networks.

- Designed and built a multi-class classification network to categorize sounds by their instruments with 76% accuracy.
- Researched state-of-the-art audio models including WaveGAN, WaveNet, and NSynth.
- Proposed methods to generate musical notes based on sound descriptors using a GAN.
- Wrote a 24-page manuscript.

WORK EXPERIENCE

06/2023 to Current

Full Stack Software Engineer

Cisco Meraki — Remote

Developing the Meraki Cloud Dashboard full-time on a global team of engineers.

- Technical Lead for the 'Power Settings' page of the Meraki Dashboard, to distribute hardware-tested, power-saving configuration options.
- Developed a distributed system for syncing Meraki manufacturing information across servers.
- Resolved several time-sensitive, million-dollar issues for clients across the globe by debugging Meraki's 4 million-line codebase.
- Worked on RAG for an in-house test generation LLM agent.
- Designed an LLM for retrieving Slack and Confluence documents to help developers access information during the Cisco hackathon.
- Performed several interviews and assisted onboarding software engineers from India and China.

01/2025 to 12/2025

Startup CTO

Beamshyft — Boston, USA

A B2B service for selling and shipping low-cost housing supplies directly from manufacturers to real estate developers. (<https://beamshyft.com>)

- Independently designed, developed, and launched a dynamic website with a video, a catalog, and a scheduling system in Next.js.
- Improved project clarity by outlining four key development phases in a statement of work.
- Adapted development phases to meet business needs.

01/2023 to 06/2023

Software Engineer Intern

Cisco Meraki — Remote

- Collaborated with the UX team to develop the updated UI for the Radio Settings page, using Cisco Magnetic design principles.
- Diagnosed high-priority issues to prevent dashboard outages for customers with large networks.

03/2021 to 02/2022

Software Engineer Intern

HouseCuts — Boston, MA

- Engineered frontend in HTML, CSS, and JS for our appointment booking system.
- Developed backend architecture with MySQL to support data management and geographical data lookup.
- Created robust API in PHP for seamless integration with Squarespace's Acuity API.

COURSEWORK

- **Machine Learning** (589, Graduate level)
- **Reinforcement Learning** (H589, Graduate level)
- **Artificial Intelligence** (383)
- **Computational Tools for ECE** (202)
- **Uncertainty, Risk, and Decision Making** (499CM)
- **Software Entrepreneurship** (420)
- **Fundamentals of Music Theory** (110)

ACTIVITIES

- Swim-team captain
- Robotics Club (computer vision)
- St. Jude Children's Research Hospital fundraising
- TKE recruitment chairman

SKILLS

AI & ML: PyTorch, Tensorflow, MediaPipe, Multimodal VAE, RAVE, VRNN, Audio/Signal Processing, CLIP, Stable Diffusion

Programming: Python, Matplotlib, Librosa, Pandas, Embedded C, CUDA, MATLAB, React/Next.js, Ruby on Rails, SQL

Fabrication & Design: CAD (Fusion 360), PCB Design (KiCAD), 3D Printing

Creative Practice: Music Production (Ableton Live), Sound Design, Adobe

Research & Interpersonal: Problem-Solving & Critical Thinking, Creativity & Innovation (Musician, Inventor, Builder), Collaboration & Communication (Research Partnerships, Recruitment Chairman), Leadership & Initiative (Swim Team Captain, Tech Lead), Adaptability & Resilience, Industriousness

LANGUAGES

English, Spanish

REFERENCES

Dr. Svetlana Boriskina: sborisk@mit.edu

Dr. Christopher Wm. White: cwmwhite@umass.edu

Tony Tsai: kunttsai@cisco.com